

values in the problem statement. Isn't that cool?

Solving geometry problems

How about finding the intersection point of two straight lines?

Let us have the following two straight lines, defined in the Cartesian coordinate system, i.e., the x-y system:

$$4x + 3y = 24$$

$$3x + 4y = 25$$

Similar to the earlier problem, the intersection point could be obtained as follows:

```
$ octave -qf
octave:1> A = [
> 4 3
> 3 4
> ];
octave:2> b = [
> 24
> 25
> ];
octave:3> X = inv(A) * b
X =
```

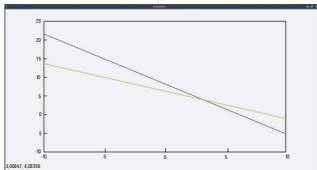


Figure 1: Intersection of straight lines

```
3.0000
4.0000
```

```
octave:4>
```

So, (3, 4) is the intersection point. If you want to see it visually, you just need to rewrite the straight line equations as follows:

$$y = (24 - 4x) / 3$$

$$y = (25 - 3x) / 4$$

And then here is the code:

```
octave:1> x=-10:0.01:10;
octave:2> plot(x, (24 - 4*x)/3, "b.", x, (25 - 3*x)/4, "g.");
octave:3>
```

Figure 1 shows the graph generated by the above code.

Solve it

Equipped with the basics of puzzle solving, here's one to tease your brain with: A vegetable seller has placed various equally priced stacks for sale at ₹ 30. One stack has 4 lemons, 7 cucumbers and 9 tomatoes. Another has 2 lemons, 5 cucumbers and 27 tomatoes. And the third has just 9 cucumbers and 15 tomatoes. Can you compute the price of each vegetable?



Hint: Assume the price of lemons, cucumber, and tomato as 'l', 'c', and 't', and then form the three equations in three variables.

If you think, you have got it, you may email the solution to me. And as we move on, we will get into some different kind of puzzle-solving. **END** 

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